

[DC] Enhancing Accessibility and User Experience of Virtual Reality Locomotion for Older Adults

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ABSTRACT

There is an increase in Virtual Reality (VR) applications geared towards Older Adults (OAs) in both medical and leisure usages. However, OAs are repeatedly excluded in technological designs, despite the significant growth of the aging population globally. This results in technological designs that are not age-friendly contributing to frustrations when trying to use VR that eventually led to non-adoption. VR locomotion is one of the many aspects that may represent barriers for the adoption of VR by OAs. VR locomotion are ways an individual may navigate within a virtual environment and is fundamental to any VR experiences. In VR research, most VR systems are designed with younger users in mind. When it comes to older users, these designs are either transferred directly or at best minimally adapted. This further suggests VR as an inaccessible platform for OAs. This PhD work aims to improve accessibility of VR experiences for OAs by addressing VR locomotion. Mixed and participatory methods will be used to focus on 1) unique OA needs in VR locomotion, 2) mental model of VR interactive mechanisms, and 3) the suitability criteria for evaluating VR locomotion techniques.

Index Terms: Virtual Reality, Older Adults, User Experience Design, Accessibility, VR Locomotion, Navigation, Interactive Mechanism, Human Computer Interaction.

1 INTRODUCTION

Virtual Reality (VR) application and research for Older Adults (OAs) have increased, such as exercise training to enhance and strengthen balance, exergame for preventing falls, and embodied environment facilitating intergenerational social interaction [7, 6, 8]. However, OAs are rarely considered as target users in technological designs. The aging population, 60 and above, is expected to double by 2050 and WHO listed the support of healthy aging for OAs as one of its initiatives [5]. The core of healthy aging is to build and maintain functional ability, and relationships, which in term explains the increased VR applications towards OAs and showcases future potential of VR supporting healthy aging [5]. Despite the increase in various VR applications, OAs often experience frustration when trying to use VR, which lead to non-adoption. One such critical frustration is that of locomotion in VR – the interaction mechanisms that allow users to move or navigate in virtual environments – and thus, a fundamental component of ensuring positive experiences when interacting with VR. Many VR locomotion studies focusing on OAs often draw knowledge from studies with younger users [2, 4, 1]. This leads to VR becoming an inaccessible interaction space for OAs, regardless of potential benefits (social, fitness etc.) it can bring to this demographic group. As such, the overall goal of my research is to contribute research and design knowledge

within the space of Human-Computer Interaction (HCI), in particular, accessibility design, by investigating how we can improve VR locomotion and experience for OAs. For this, my PhD work addresses the following research questions:

- RQ1) What are older users' preferences and restrictions that impact user experience with existing VR locomotion techniques?
- RQ2) How can we measure subjective VR experiences and suitability of VR locomotion techniques for OAs?
- RQ3) How can we enhance the accessibility of VR locomotion techniques for OAs?

2 RESEARCH PROGRESS

My PhD research is structured along three phases to address the RQs: 1) addressing the lack of VR locomotion technique taxonomy for OAs 2) identifying the suitability of VR locomotion techniques for OAs 3) evaluating and co-designing VR interaction mechanism. I am currently halfway through phase 1 where I have completed a scoping review. I have recently obtained my ethics approval for Study 1 and started the planning process for Study 2.1.

2.1 Scoping Review for VR Locomotion Techniques

In the first year of my PhD, I conducted a scoping review, with the PRISMA method, of head-mounted display (HMD) VR locomotion techniques, focusing on translational and rotational locomotion [3]. The review produced a holistically understanding of the VR locomotion research landscape for OAs. In addition, I explored how suitability was evaluated in every locomotion technique paper. Suitability, in this context, refers to how appropriate a VR locomotion technique is for OAs in specific applications or situations. The search was conducted on The ACM Library and IEEE Xplore databases to encompass cross-disciplinary collaboration and newest technological explorations. Empirical study papers with direct OAs participation or mentions of OAs as main users were included. A total of 399 papers were identified from both databases. After screening and reviewing, 47 papers were included for the meta-analysis.

The results showed that motion-based locomotion techniques were the most popular in research [3]. However, the lack of positive feedback, and fear of moving freely and falling were common. This contributed to OAs feeling unconfident in using VR independently and restricting locomotion in VR studies to minimize disorientation and discomfort. Suitability was generated from task performance with every VR locomotion technique. Although the analysis of suitability was efficient for the scope of this review, it would be difficult to use the same analysis for less studied applications such as social VR, simulation and entertainment, where there would be no clear performance differences. This scoping review was presented and published as a workshop paper at IEEE VR 2025. The findings guided planning for the three phases of this PhD work.

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2.2 Evaluating Existing Locomotion Techniques

This study (Study 1) is grounded in the findings from the scoping review and has received ethics approval at the university. The objective is to co-evaluate common VR locomotion techniques with OAs in a mixed factorial, randomized trial study. The study is separated into two parts where the first study will focus on evaluating VR locomotion technique in a seated position, while the second study is in a standing position. OA participants will be asked to perform simple navigation tasks with different VR locomotion techniques. The VR environment used is a mansion with multiple rooms filled with object seeking tasks. Task difficulty and cognitive load of each task are controlled in every room. I am currently planning for pilot studies to refine the study method while initiating the recruitment of OA participants with our long-term care home partners as recruitment is expected to be longer due to the inclusion and exclusion criteria. I expect the results of this study will inform the development of measuring suitability and uncover important aspects of VR experiences for OAs.

2.3 Taxonomy and Determining a Suitability Scale

I am in the planning phase for Study 2, and it is also grounded from the scoping review results. The goal is to conduct a meta-analysis of all existing VR locomotion technique taxonomies to construct and propose a taxonomy specifically for OAs. As a first step, I am planning to revise all VR locomotion technique taxonomies and organize it, either merge or keep separate, in a way that can be used for an expert review on evaluating commercial VR applications (Study 2.1). The results of Study 1 and Study 2.1 will inform a working list of criteria for assessing VR locomotion suitability for OAs, which will be further evaluated in phase 3.

2.4 Future Studies

Phase 3 includes exploring alternatives or modifications of VR locomotion interaction mechanisms (both hardware and software) for OAs. For instance, I plan to redesign handheld controllers based on the usability and accessibility issues uncovered from previous phases. I will be utilizing quick prototyping techniques, like 3D printing, to create handheld controller prototypes, and Unity to map interactions. Prototypes will be tested and iterated, at least 2 rounds, with OAs through participatory design workshops to ensure they are meaningfully involved in all stages before moving to the final evaluation stage. The final evaluation stage will be a replication of Study 1 with the new locomotion technique added as an independent variable that will be evaluated against other existing techniques.

3 CONCLUSION AND FEEDBACK

This PhD work addresses the lack of understanding and inaccessibility of VR experiences for OAs through the lens of VR locomotion. The contributions of this work include enriching the literature by filling in the research gap, addressing accessibility barriers to increase adoption of VR by OAs, and acting as an anchor for designers or developers when creating VR experience for OAs.

Considering the scope of this PhD work, I would like feedback and comment on the following questions:

- How can we incorporate sociotechnological aspects such as accessibility considerations and personal factors into engineering-based design work in order to make VR experiences more age-inclusive?
- How can we measure subjective VR experiences and enjoyment for underrepresented population that have significant accessibility barrier in using VR?

ACKNOWLEDGMENTS

The author wishes to acknowledge the financial endowment received through the Schlegel Research Chair in Technology for Healthy Aging held by Prof. Cosmin Munteanu.

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